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Roll No

FT-705-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Chemical Safety

Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. a) Give the detail description on human lungs structure? Explain the process of Pneumoconiosis with primary and secondary stages.
b) Explain the dynamics and kinematics of toxicology in human body? Explain the general principles of exposure and response?
2. Explain the mechanism of BLEVES with different hazardous characteristics. Explain Hymes's point source model and Robert's model for radiation intensity and the maximum diameter of the fire ball respectively.
3. a) Enlist at least ten chemical substances with target organ and occupation settings having potential to produce human carcinogenicity?
b) Define the toxicity classes based on LD₅₀ and LC₅₀ for the toxic materials available in occupational settings?

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4.
 - a) Define flammability? Explain its limit and how this limit helps to evaluate degree of hazard.
 - b) Define Toxicity and give in brief the quantitative estimation method.
5. What is the dust explosion? Explain the mechanism of dust explosion.
6. On an overcast day stack with an effective height of 60 m is releasing sulfur dioxide at the rate of 80 g/s. The wind speed is 6 m/s. The stack is located in rural area determine.
 - a) The mean concentration of SO₂ on ground 500 m downwind?
 - b) The mean concentration on ground 500 m downwind and 50 m cross wind?

Data

$$6y = 0.08x (1+0.0001x)^{-1/2}$$
$$6z = 0.06x (1+0.0015x)^{-1/2}$$
7.
 - a) Enlist the details description on input data and output data for Risk Analysis Methodology available for System Safety Assessment.
 - b) Define probabilistic deterministic and qualitative and Quantitative approaches with their individual risk matrix of consequences scales.
8. Explain Flash Fire, jet fire and pool fire on the following point.
 - a) Flame temperature
 - b) Flame shape
 - c) Duration
 - d) Volume and area

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